

Day-Ahead Electricity Price Forecasting

A Probabilistic LASSO-Based Approach
for the Germany-Luxembourg Electricity Market

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Data Overview

92,519

Hourly Records

Jan 2015 – Jul 2025

46

Features

After engineering

2,363

Negative Prices

–500 to +936

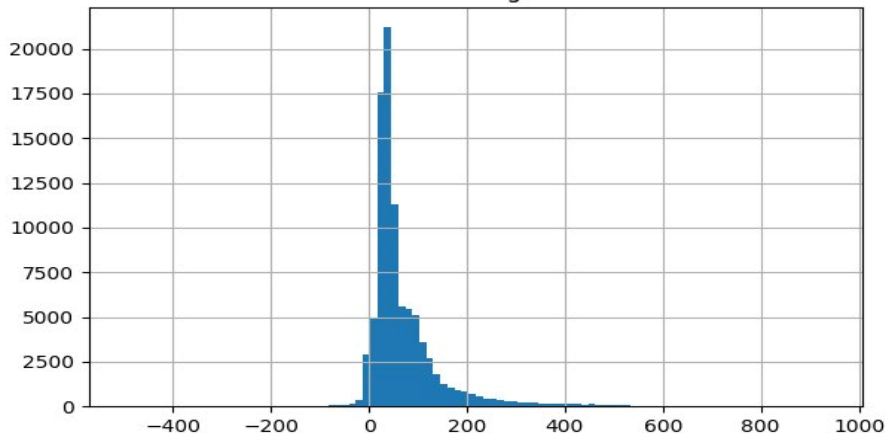
Price Range

€/MWh

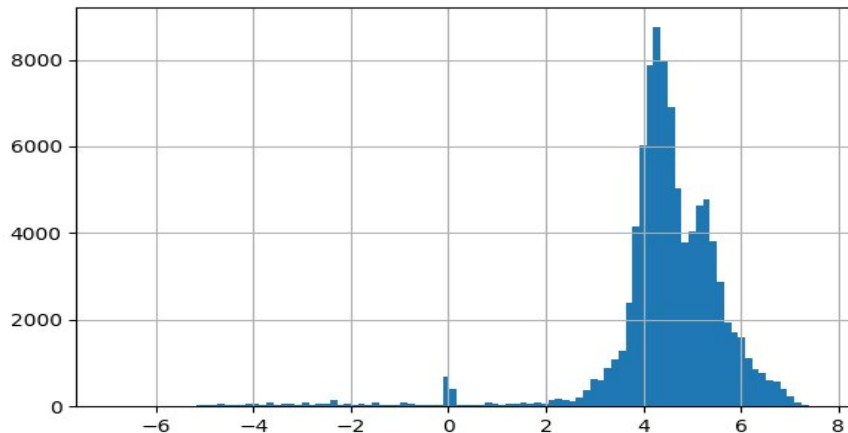
Forward-fill for missing commodities

Dropped Actuals and Temp_DA (59% missing)

Price - Original



Price - asinh transformed



Price distribution - asinh transformation handles negative prices and reduces skewness.

Feature Engineering

Price Lags

- lag_1 — price one hour ago (strongest signal)
- lag_{24} — same hour yesterday
- lag_{168} — same hour last week
- lag_{48} — two days ago

Market Variables

- $\text{Residual Load} = \text{Load} - \text{Solar} - \text{Windon} - \text{Windoff}$
- **Marginal Cost Indicators (Commodities):**
- Gas, Coal, Oil, EUA, EUR/USD

Calendar Dummies

- 23 hour of day dummies (drop h_0)
- 6 weekday dummies (drop Monday)
- One-hot encoding

Target Transformation

- $y = \text{arcsinh}(\text{Price})$ — handles negative prices

Multivariate LEAR: LASSO-Estimated AutoRegressive Model

LASSO regression for feature selection and prediction

0.000409

Best λ (alpha)

Tuned alpha via TimeSeriesSplit CV

40 / 46

Features Selected

LASSO removed 6 irrelevant

2015–2022

Train Period

69,865 hourly observations

2023

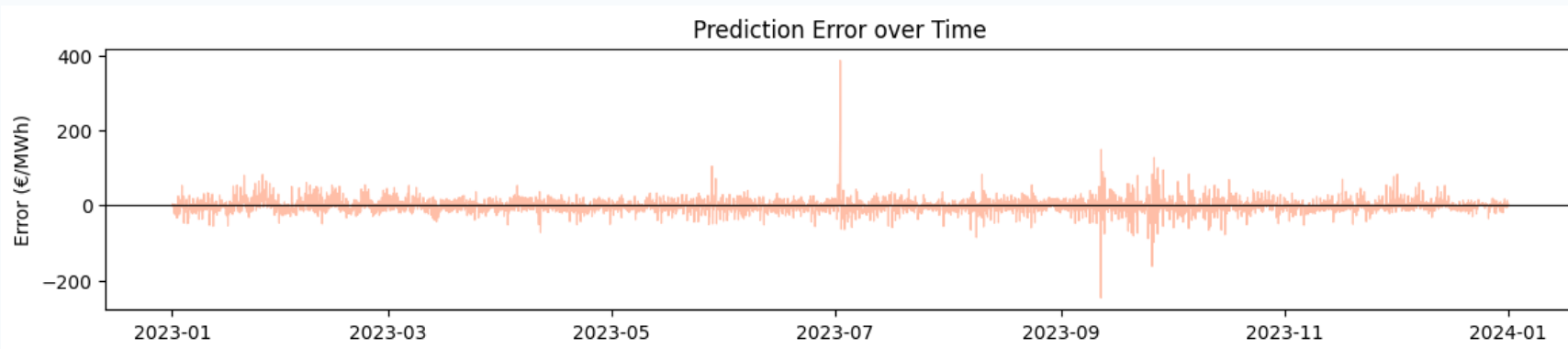
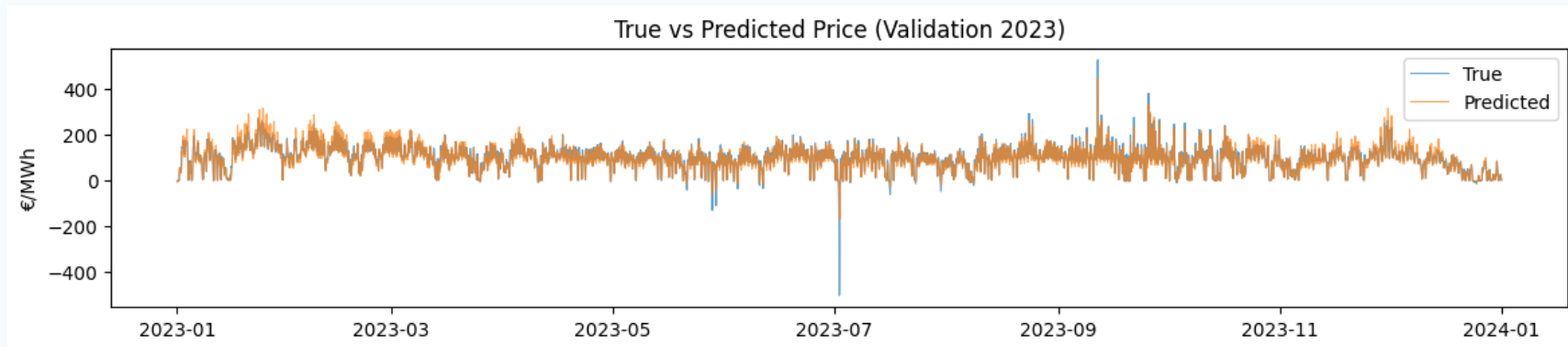
Valid Period

8,760 hourly observations

Key Findings:

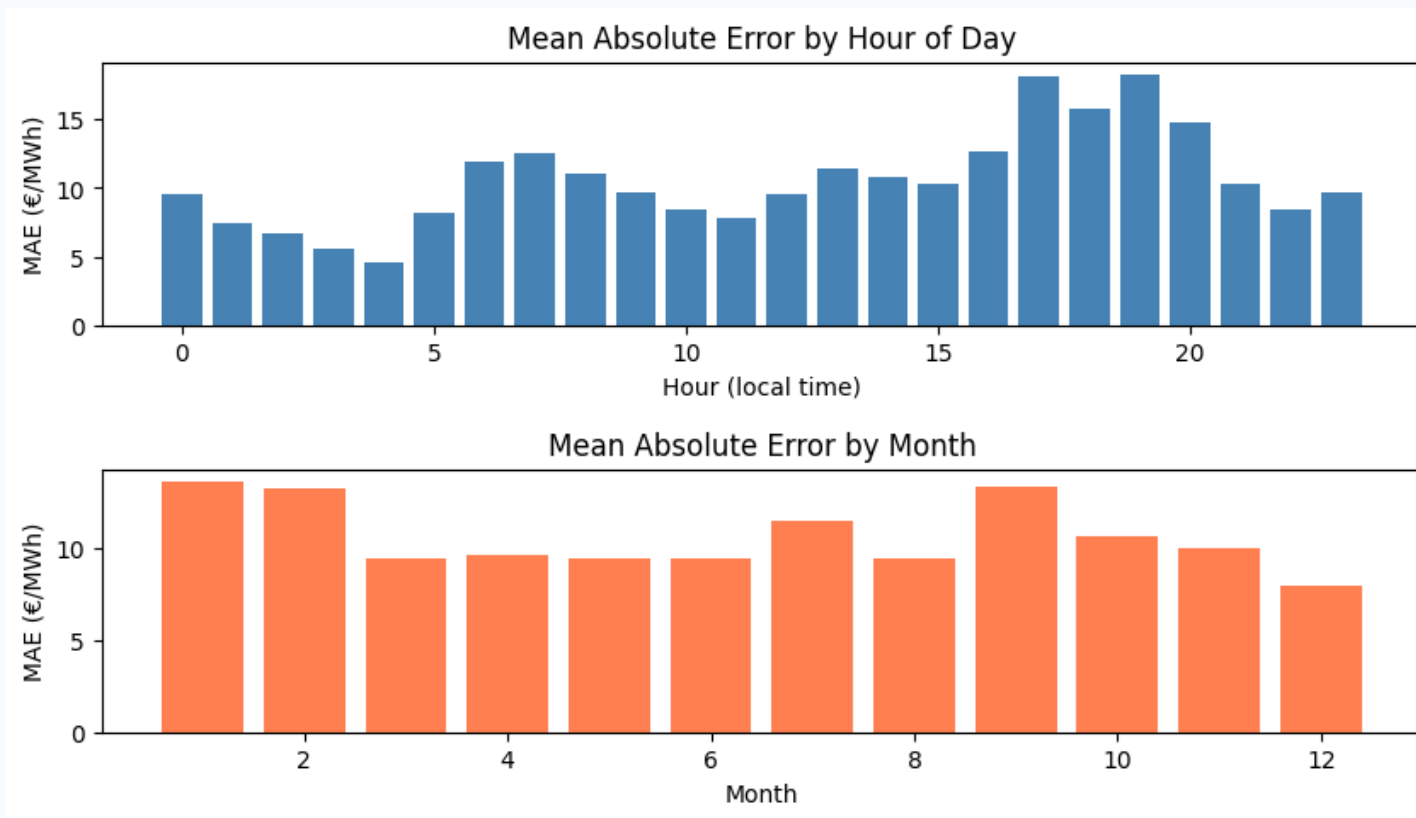
- Lag₁ dominates (coeff = 1.12)
- Removing lag₁ increases error 3.5×
- Residual_Load captures supply-demand balance

Error Analysis



July 2023: Price = -500 €/MWh (extreme solar generation-oversupply)

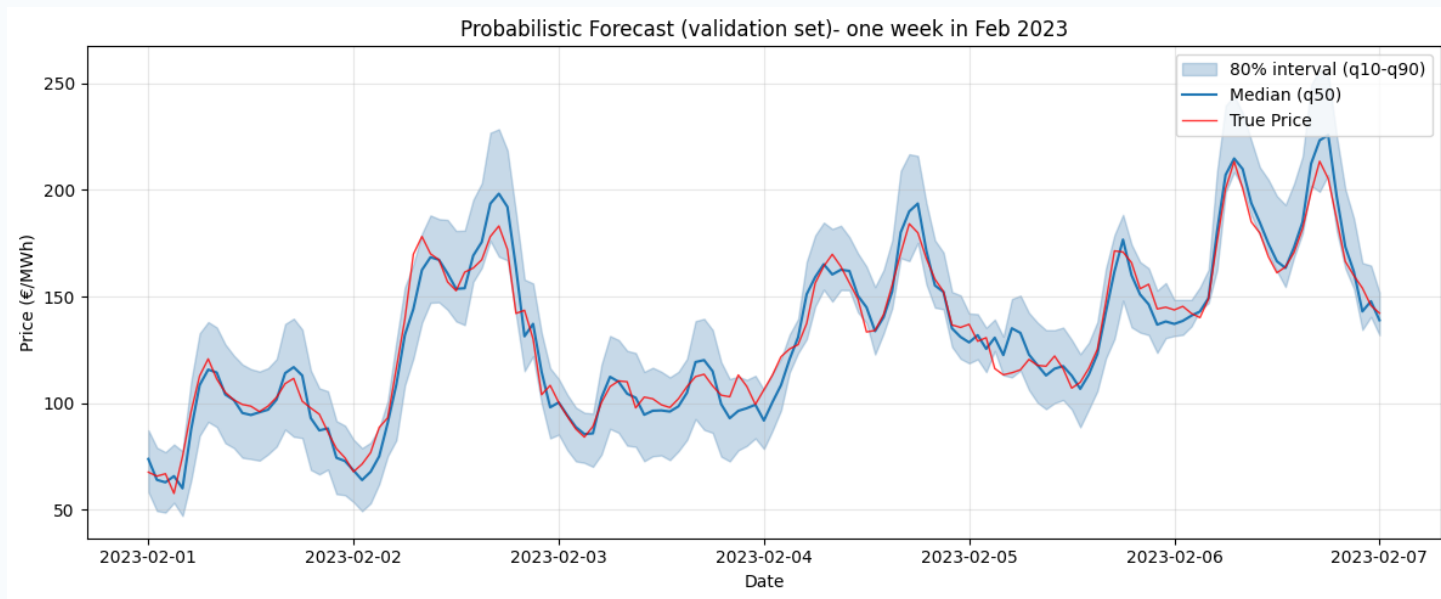
Error Analysis



Peak errors at 17–20h · Worst months: Jan, Feb, Sep

Probabilistic Forecast — Quantile Regression

Model Tuning & Validation - Year 2023



74.9%

Validation Coverage
(q10-q90)

Expected 80%

$q_{10}, q_{90} = 0.0063$

Best λ (alpha)

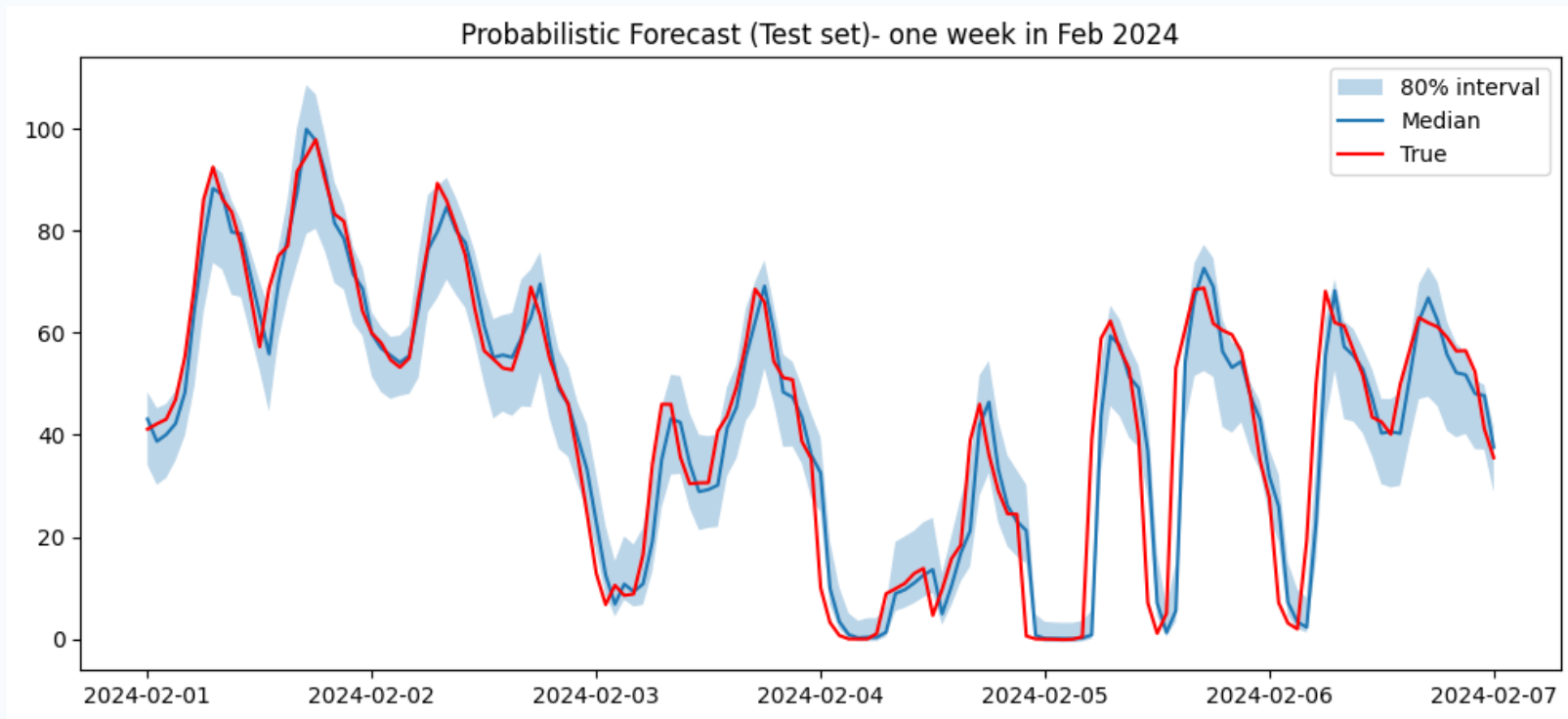
Tuned $\lambda_{10,90}$ via Pinball Loss Minimization

4.06

Pinball Loss
(q50)

Probabilistic Forecast — Quantile Regression

Final Testing- Year 2024



Results

Metric	Validation	Test
MAE	10.58 €/MWh	9.97 €/MWh
RMSE	16.21 €/MWh	18.46 €/MWh
Coverage (q10–q90)	74.9%	56.4%

Future Work:

- Use non-linear models
- Combine models using ensemble approaches
- Improve uncertainty estimation
- Better handle extreme price events

The MAE is higher than benchmark results in the literature, which can be explained by higher volatility in recent years and a simpler model setup.

The coverage is low because the model underestimates uncertainty, the prediction intervals are too narrow, especially during extreme price events.